

12. Future Improvements

Although this project provides valuable insights into Airbnb pricing, demand patterns, and investment opportunities, several improvements can be implemented to enhance the accuracy, scalability, and business value of the analysis.

1. Incorporate Additional Data Sources

The current analysis relies mainly on Airbnb listing information and review data. While these datasets provide valuable insights into market behaviour, additional external data sources could improve the accuracy of investment recommendations. Future analysis could integrate:

- Tourism data (visitor arrivals, tourist spending, seasonal travel patterns)
- Local economic indicators (employment growth, population changes, income levels)
- Real estate market data (property purchase prices, rental yields, mortgage rates)
- Transportation accessibility data (distance to airports, public transport availability)
- Local event data (festivals, conferences, major events)

2. Improve Revenue Prediction Through Additional Variables

The current machine learning model focuses mainly on predicting listing prices. However, price alone does not fully represent investment profitability. A higher-priced property does not necessarily generate higher returns if occupancy is low. Future models should incorporate revenue-related metrics such as occupancy rate, booking frequency, average length of stay, seasonal demand patterns, operating expenses, and property acquisition costs. Besides, the next project can add a more comprehensive model could estimate investment return, which allows investors to evaluate profitability and return on investment (ROI) rather than only predicting market prices.

$$\text{Investment Return} = \text{Revenue Generated} - \text{Property and Operating Costs}$$

3. Develop a More Advanced Investment Recommendation Model

The current dashboard provides comparisons between cities, neighbourhoods, and property characteristics. However, future work could develop an automated investment scoring system which can transform the analysis from a descriptive dashboard into a decision-support system that directly recommends the most attractive investment opportunities by combining predicted listing price, demand intensity, review performance, competition level, location attractiveness, and revenue opportunity. Each property could receive an overall investment score, allowing investors to quickly compare opportunities

4. Improve Machine Learning Model Performance

Although the Random Forest model achieved strong performance (R^2 approximately 0.75), further improvements could be explored.

Additional features could be created, such as:

- Distance to city centre
- Distance to tourist attractions
- Number of nearby amenities
- Price per accommodation capacity
- Neighbourhood popularity score
- Review growth rate

Hyperparameter Optimisation

- Future models could apply grid search, random search, or Bayesian optimisation to identify the optimal model configuration.

5. Enhance Dashboard Functionality

Future dashboard improvements could include the interactive investment simulator, in which users could input target city, budget, property type, and number of bedrooms to receive recommended investment opportunities.

Besides, the dashboards can also integrate with live Airbnb or market data, which allows continuous monitoring, updated pricing recommendations, and faster decision-making.